

Asy-Syahid Abdurrahman Hanan Taqiyya

Software Engineer (Full-Stack & Systems)

hanan7taqiyya@gmail.com • +62 851-5783-9155 • github.com/VUXXE • Jakarta, Indonesia

SUMMARY

Informatics undergraduate and Software Engineer focused on full-stack product engineering, edge architectures, and real-time systems. Experienced in React/TanStack Start apps on Cloudflare (D1/R2), memory-efficient desktop apps in Rust/Tauri & Java, messaging bridges in Node.js/Baileys, and clean-architecture APIs in Go. Passionate about building low-overhead, local-first software and production-grade tools.

EDUCATION

Universitas Indraprasta PGRI (UNINDRA) — Jakarta, Indonesia

Bachelor of Science in Informatics Engineering (Teknik Informatika) | **GPA:** 3.49 / 4.00 | 2023 – Present (Expected 2027)

Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Computer Architecture, Compiler Design, Object-Oriented Programming

TECHNICAL SKILLS

- **Languages:** Go, Rust, TypeScript, JavaScript, Java, Python, SQL, HTML5/CSS3
- **Frameworks & Libraries:** React 19, Next.js, TanStack Start, SvelteKit (Svelte 5 Runes), Go Fiber v2, GORM, Tauri v2, FlatLaf
- **Databases & Storage:** PostgreSQL, MySQL 8.0, Cloudflare D1/SQLite, Cloudflare R2, Drizzle ORM, HikariCP
- **Cloud & Infrastructure:** Cloudflare Workers, Durable Objects, Docker, Linux (Arch/CachyOS), Git/GitHub Actions, OCI
- **Concepts & Architecture:** Clean Architecture, Better Auth, Real-Time State Sync (LWW), Baileys/WhatsApp APIs, OS System Integration, RESTful APIs

KEY TECHNICAL PROJECTS

Mesh — Real-Time Collaborative Vector Whiteboard

Technologies: TypeScript, SvelteKit (Svelte 5), Cloudflare Workers, Durable Objects, SQLite, Docker

- Engineered an edge-native, serverless vector whiteboard deployed on Cloudflare Workers with zero reverse proxy or CORS overhead.
- Implemented distributed room coordination via Durable Objects with embedded SQLite persistence and monotonic Last-Write-Wins (LWW) conflict resolution.
- Developed a dual-layer HTML5 canvas engine with a committed static buffer and 60 FPS interactive overlay featuring RDP path smoothing.

WhatsApp Tauri — High-Performance Desktop Client

Technologies: Rust, Tauri v2, Webview, Linux/Windows/macOS APIs

- Built a lightweight, cross-platform WhatsApp desktop application with Tauri v2 and Rust, reducing binary size to ~5.7MB (30x smaller than Electron).
- Cut memory footprint to ~120MB baseline RAM and achieved zero background idle CPU usage by offloading to native OS webview event loops.
- Integrated system-level notifications directly with OS daemons (libnotify on Linux, Windows Toast, macOS NSUserNotificationCenter).

Rakamin Evermos — Clean Architecture E-Commerce API

Technologies: Go (1.25+), Fiber v2, MySQL 8.0, GORM, Docker, JWT

- Architected a scalable e-commerce RESTful API in Go adhering to Clean Architecture and strict multi-tenant domain boundaries.
- Implemented atomic checkout transactions with pessimistic row-locking (SELECT FOR UPDATE) to eliminate inventory race conditions.
- Delivered JWT authentication, RBAC middleware, and administrative regional lookup with in-memory caching and 100% test coverage.

Perpustakaan Freedom — Desktop Library Management System

Technologies: Java 21, Java Swing, FlatLaf, MySQL 8.0, HikariCP, JasperReports, BCrypt

- Led core engineering for a modern desktop library system (Group Project) featuring automated circulation, fine calculation, and BCrypt security.
- Integrated HikariCP high-performance connection pooling, dynamic JasperReports generation, and JFreeChart analytics dashboards.
- Engineered an offline-ready Maven configuration with bundled local repository dependencies for zero-network compilation.

Baswara — Digital Invitation Builder (Cloudflare Edition)

Technologies: TypeScript, TanStack Start (React 19), Cloudflare Workers, D1, R2, Drizzle ORM, Better Auth

- Built a digital invitation platform (visual builder, RSVP tracking, guest check-in) 100% on Cloudflare Workers with D1 + R2.
- Implemented Better Auth email/password sessions in D1, R2 asset uploads, and dynamic OG-image/QR generator endpoints.

WhatsApp Bridge Custom — Baileys Group Management API

Technologies: Node.js 20+, Baileys v6, REST, WhatsApp Web Protocol

- Extended a Baileys-based WhatsApp bridge with custom group endpoints: list, invite links, create-group, add-member, promote, update-description.
- Added hardened send/media/poll/location/edit/typing/read APIs with bot/self-chat modes and allowlist-based access control.